

ENDGAMES

STATISTICAL QUESTION

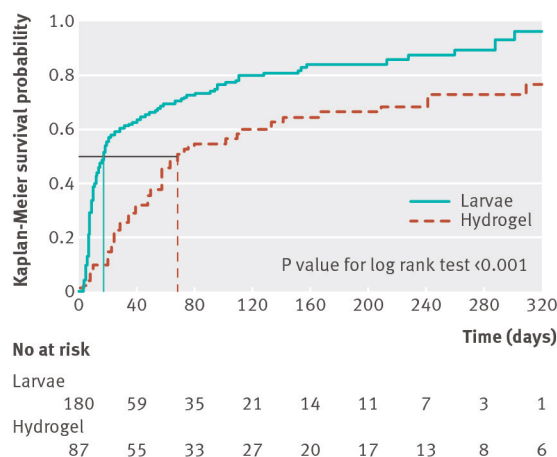
How to read a Kaplan-Meier survival plot

Philip Sedgwick *reader in medical statistics and medical education*

Institute for Medical and Biomedical Education, St George's, University of London, London, UK

Researchers investigated the effectiveness of larval therapy for the treatment of sloughy or necrotic leg ulcers. A three armed randomised controlled trial study design was used. The intervention consisted of loose larvae or bagged larvae. The control treatment was the standard debridement technique of hydrogel. Participants had at least one venous or mixed venous and arterial ulcer, with at least 25% coverage of slough or necrotic tissue. The outcome measures included time to healing of the largest eligible ulcer and time to debridement (defined as a cosmetically clean wound) after the ulcer healed. The maximum length of follow-up was 12 months.¹

In total, 267 participants were recruited and randomised to loose larvae (n=94), bagged larvae (n=86), or hydrogel (n=87). Healing rates did not differ between the loose and the bagged larvae arms, so the results were presented for larvae overall (loose and bagged larvae arms combined). Time to healing was not significantly different between the larvae groups (combined) and hydrogel treatment groups (hazard ratio for healing using larvae v hydrogel 1.13, 95% confidence interval 0.76 to 1.68; P=0.54). Larval therapy significantly reduced the time to debridement compared with hydrogel (hazard ratio 2.31, 1.65 to 3.2; P<0.001; figure). It was concluded that larval therapy did not improve the rate of healing of sloughy or necrotic leg ulcers compared with hydrogel but did significantly reduce the time to debridement.



Kaplan-Meier plot of time to debridement of ulcer using larval therapy (loose and bagged combined) compared with hydrogel. The median survival time for each treatment group is the length of time corresponding to the probability of 0.5

Which of the following statements, if any, are true?

- If a patient did not experience debridement during follow-up, their observation would be "censored"
- Participants were excluded from the analyses if they did not experience debridement before the end of follow-up
- The curves did not achieve a probability of 1.0 because some observations were censored at the end of follow-up
- The log rank test provides an estimate of the magnitude of difference in times until debridement between treatment groups
- The probability estimates decreased in accuracy as fewer participants became at risk of debridement

Answers

Statements *a*, *c*, and *e* are true, whereas *b* and *d* are false.

The trial investigated the effectiveness of larval therapy for the treatment of sloughy or necrotic leg ulcers. A randomised

controlled trial study design was used. For the purpose of analysis, the loose larval and bagged larvae groups were combined. Outcomes included the time after randomisation until healing and time until debridement after healing. The times until healing and debridement are referred to as “survival data.” More generally, survival data comprise the time it takes a patient to reach an endpoint. The term survival data is possibly misleading because the endpoint does not have to be death or an adverse event. The endpoint might be positive as in the trial above. Survival data are often referred to as “time to event data.”

If a participant experienced debridement during follow-up, his or her survival time was referred to as “exact.” The distinguishing feature of time to event data is that some participants do not experience the endpoint before the end of the study. In the above trial, for those participants who had not experienced debridement during follow-up, their survival time was recorded as the length of follow-up. It is not known exactly when they experienced debridement, except that it occurred sometime after the study finished. For that reason, their survival times are called “right censored” or simply “censored” (*a* is true). Censored observations would have occurred for two main reasons. A participant may have been followed for the maximum duration of follow-up and not experienced debridement during that time. Secondly, for participants who were randomised to treatment towards the end of recruitment, the length of follow-up may not have been long enough to experience debridement. Some participants may also have had censored observations because they withdrew from the study or were lost to follow-up before experiencing debridement. Types of censoring for survival data have been described in a previous question.² Participants with censored data provide valuable information about the effects of treatment and are therefore still included in the analyses (*b* is false).

Survival times are represented graphically by survival curves, calculated by the Kaplan-Meier method. For the above trial, the probability of debridement was plotted against the duration of follow-up for each treatment group (figure). The method assumed that censoring was not related to the time taken to achieve debridement. The survival curves are described as “plotted going up.” Time zero on the plot represents the healing of the largest eligible ulcer. At that stage, none of the participants had yet experienced debridement and therefore the probability of debridement was equal to zero. The curves increased in probability only when a trial participant experienced debridement. The curves are therefore represented as a step function. Censored observations are still included when calculating the probabilities. However, the survival curves do not change in probability when an observation is censored but when the next subsequent exact survival time occurs. The probabilities, so called Kaplan-Meier probabilities, are cumulative ones and therefore show the proportion of the treatment groups that had experienced debridement at each time point, or earlier, during follow-up.

Neither survival curve reached a probability of 1.0 by the end of follow-up. This was because both groups contained participants with censored observations in (*c* is true). These participants took longer than the maximum length of follow-up to achieve debridement. However, if the longest observed survival time was exact rather than censored, the survival curves would have reached a probability of 1.0 by the end of follow-up. Furthermore, if none of the times until debridement had been censored, the Kaplan-Meier probabilities would have been equivalent to absolute ones.

The median survival time is of particular importance when describing time to event data. It is the duration of follow-up at

which the Kaplan-Meier probability is 0.5 and it represents the average survival time. The median survival time until debridement is derived by drawing a horizontal line across from the probability of 0.5, and it is the time point at which this line crosses the curve for a treatment group (figure). The hydrogel group had a median time until debridement of 72 days (95% confidence interval 56 to 131 days). The confidence interval is relatively wide, which indicates that the sample estimate was not an accurate one. The researchers did not report the median time until debridement for the larvae treatment groups combined. Using the Kaplan-Meier plot (figure), it is estimated that the median time until debridement was about 18 days. Therefore, half of the combined larvae treatment group had experienced debridement within about 18 days after healing of the ulcer (or sooner) compared with 72 days for the hydrogel group.

The treatment groups were compared in the time until debridement using the log rank test. This test has been described in a previous question,³ and it provides a general comparison of the survival times for treatment groups during follow-up. It is based on traditional statistical hypothesis testing, with a null and alternative hypothesis, as described in a previous question.⁴ The null hypothesis starts at the position of equipoise, and states that there is no difference in debridement times between participants receiving larval treatment and hydrogel in the population from which the sample was obtained. The alternative hypothesis states that a difference exists—that is, in the population from which the sample was obtained the debridement times for the treatment groups are not equal. The resulting *P* value was less than 0.001, and because it was less than 0.05 the null hypothesis was rejected in favour of the alternative. It was concluded that the trial provided evidence of a difference between treatment groups in the time until debridement. The curve for the larvae group was above that for the hydrogel group (figure), and it can therefore be inferred that times until debridement were shorter in the larvae group than in the hydrogel group.

The log rank test is purely a test of significance. It does not provide an estimate of the magnitude of difference in survival times between treatment groups (*d* is false). The hazard ratio, described in a previous question,⁵ is often derived to provide such an estimate of the difference and provides a measure of relative survival times between treatment groups. For the above trial, the hazard ratio for the time until debridement for combined larval therapy versus hydrogel was 2.31 (1.65 to 3.2). The hazard ratio is an estimate of the rate, sometimes referred to as instantaneous risk, of debridement in the larvae group relative to the hydrogel group at any time during follow-up. The use of the word “risk” is perhaps misleading—it is used in this context to indicate probability and not the potential for an adverse outcome. At any time during follow-up, those in the larvae group were 2.31 times as likely—that is, they had an increased risk of 131%—of achieving debridement compared with the hydrogel group. This increase in risk was significant because the 95% confidence for the hazard ratio did not include unity (1.0), which would indicate equal rates of debridement between treatment groups. Although the confidence interval was wide—representing an increased instantaneous risk of between 65% and 220%—the lower limit was probably sufficiently far enough away from unity to reflect a clinically important increased risk. The derivation of the hazard ratio assumed that the relative rates of achievement of debridement, and therefore their ratio, did not vary greatly during follow-up. However, if the survival curves had crossed, the assumption would not hold true and the hazard ratio would not have been a useful measure. It is not possible to predict the time until

debridement from the hazard ratio, only that the larvae group achieved it significantly more quickly than the hydrogel group. The significant difference between treatments in time until debridement described above is apparent in the Kaplan-Meier survival plot. The survival curves for both treatments increased sharply at the start of follow-up. After 80 days, the Kaplan-Meier probabilities were about 0.75 for larval therapy and 0.55 for hydrogel. Thereafter, the curves started to flatten out and at the end of follow-up the probabilities were about 0.95 and 0.78 for the larval and hydrogel treatment groups respectively.

The numbers of patients in each group that were at risk of debridement during follow-up—that is, had not achieved debridement—are shown under the Kaplan-Meier survival plot. Towards the end of the study these numbers dropped. Therefore, the Kaplan-Meier survival probabilities would have decreased in accuracy because they were based on decreasing numbers of participants (*e* is true). As is typical of most Kaplan-Meier plots, the curves for the trial above flattened out on the right hand side as fewer participants achieved debridement. Caution is generally needed not to over interpret this part of the plot unless a reasonable number of subjects are still at risk.

Kaplan-Meier plots are often “plotted going down.” For the above trial, this would have entailed plotting the proportion of participants that had not experienced debridement against length of follow-up. If plotted going down the survival curve would start at the probability of 1.0 and decline with time. The probabilities would still be cumulative ones and would in principle contain the same information as a curve plotted going up. However, because the time until debridement incorporated censored data, a curve plotted going down would be interpreted slightly differently. For such curves, the Kaplan-Meier probabilities would represent the probability of experiencing debridement so many days, or longer, after starting treatment. Competing interests: None declared.

- 1 Dumville JC, Worthy G, Bland JM, Cullum N, Dowson C, Iglesias C, et al; on behalf of the VenUS II team. Larval therapy for leg ulcers (VenUS II): randomised controlled trial. *BMJ* 2009;338:b773.
- 2 Sedgwick P. Kaplan-Meier survival analysis: types of censored observations. *BMJ* 2013;347:f4663.
- 3 Sedgwick P. The log rank test. *BMJ* 2010;341:c3773.
- 4 Sedgwick P. Understanding statistical hypothesis testing. *BMJ* 2014;348:g3557.
- 5 Sedgwick P. Hazards and hazard ratios. *BMJ* 2012;345:e5980.

Cite this as: *BMJ* 2014;349:g5608

© BMJ Publishing Group Ltd 2014